HU Collection

Test of Mathematics for University Admission

Mock Set D Paper 1

Applications of Mathematical Knowledge

Time allowed:	**75 minutes**
Questions:	20 multiple choice
Calculators:	not permitted
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Date written:	24 September 2026

Name: _____

Date sat: _____

Score: _____ / 20

Instructions

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Every question uses AS content only. The difficulty lies in how that content is used: parameters that change the case, conditions that quietly remove a solution, and answers that come from structure rather than from brute force.

Q	1	2	3	4	5	6	7	8	9	10
Ans										
Q	11	12	13	14	15	16	17	18	19	20
Ans										

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Question 1

Real numbers a, b and c satisfy

$$4^a = 9, \qquad 27^b = 25, \qquad 125^c = 64.$$

Find the value of abc.

A $\frac{2}{3}$
B 1
C $\frac{4}{3}$
D $\frac{3}{2}$
E 2
F $\frac{8}{3}$

Question 2

The curve $y = f(x)$ has exactly two stationary points: a local minimum at $(2, -3)$ and a local maximum at $(-1, 4)$. Which of the following is a local maximum point of the curve

$$y = 3 - 2f(4 - 2x)\,?$$

A $\left(\frac{5}{2}, -5\right)$
B $(1, 9)$
C $(1, -3)$
D $(-1, 9)$
E $\left(\frac{5}{2}, 11\right)$
F $(3, 9)$

Question 3

Find the coefficient of x^4 in the expansion of

$$(2 + x)^6 \left(1 - \tfrac{x}{2}\right)^6.$$

A 0
B 15
C 30
D 60
E 120
F 240

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Question 4

An arithmetic progression has first term a and non-zero common difference d. Its 1st, 4th and 13th terms, in that order, are the first three terms of a geometric progression, and the sum of the first 20 terms of the arithmetic progression is 440. Find the sum of the first five terms of the geometric progression.

A 121
B 186
C 242
D 324
E 363
F 484

Question 5

For how many real values of a does the equation

$$x^3 - (a+1)x^2 + (a+1)x - 1 = 0$$

have a repeated root?

A 1
B 2
C 3
D 4
E 5
F no values

Question 6

How many solutions does the equation

$$\tan x = 2\sin x$$

have in the interval $-360° \leq x \leq 360°$?

A 4
B 5
C 7
D 8
E 9
F 11

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Question 7

Find the complete set of real values of x for which

$$\log_2 x + \log_x 16 < 5.$$

A $0 < x < 1$ or $2 < x < 16$
B $2 < x < 16$
C $x < 1$ or $2 < x < 16$
D $1 < x < 16$
E $0 < x < 2$ or $x > 16$
F $0 < x < \frac{1}{2}$ or $2 < x < 16$

Question 8

Find the complete set of real values of x for which

$$\frac{2x}{x-1} \geq x.$$

A $0 \leq x \leq 3$
B $1 < x \leq 3$
C $x \leq 0$ or $x \geq 3$
D $x \leq 0$ or $1 \leq x \leq 3$
E $x < 1$ or $x \geq 3$
F $x \leq 0$ or $1 < x \leq 3$

Question 9

A triangle has sides of length x, $x+1$ and $x+2$, and one of its angles is $120°$. Find its area.

A $\frac{15\sqrt{3}}{16}$
B $\frac{15\sqrt{3}}{8}$
C $\frac{3\sqrt{3}}{4}$
D $\frac{15}{8}$
E $\frac{35\sqrt{3}}{16}$
F $\frac{5\sqrt{3}}{4}$

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Question 10

Find the complete set of real values of k for which the equation

$$9^x - (k+1)\,3^x + k = 0$$

has exactly one real solution.

A $k = 1$
B $k \leq 0$
C $k \leq 0$ or $k = 1$
D $k < 0$ or $k = 1$
E $k \geq 1$
F $0 < k \leq 1$

Question 11

Find the complete set of real values of m for which the line $y = mx$ meets the curve $y = x^3 - 3x^2 + 2x$ at exactly three distinct points.

A $m > -\frac{1}{4}$
B $m \geq -\frac{1}{4}$
C $m > -\frac{1}{4}$ and $m \neq 0$
D $m > -\frac{1}{4}$ and $m \neq 2$
E $m > 2$
F $m < -\frac{1}{4}$

Question 12

The circle C has centre $(3, 4)$ and passes through the origin O. How many lines through O meet C again at a point P with $OP = 8$?

A 0
B 1
C 2
D 3
E 4
F infinitely many

Question 13

Find the total area of the regions enclosed between the curve $y = x^3 - 4x$ and the x-axis for $-2 \leq x \leq 3$.

A $\frac{25}{4}$
B $\frac{33}{4}$
C $\frac{41}{4}$
D $\frac{49}{4}$
E $\frac{57}{4}$
F $\frac{65}{4}$

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Question 14

In triangle ABC, $AB = 10$, $BC = 6$ and angle $BAC = 30°$. There are two possible triangles. Find the positive difference between their areas.

A $2\sqrt{11}$
B $5\sqrt{3}$
C $10\sqrt{3}$
D $5\sqrt{11}$
E $10\sqrt{11}$
F 0: the triangle is determined

Question 15

How many tangents to the curve $y = x^3 - 3x$ pass through the point $(2, 2)$?

A 1
B 2
C 3
D 4
E 5
F 6

Question 16

Find the complete set of real values of k for which the curve

$$y = x^4 - 4x^3 + kx^2$$

has exactly one stationary point.

A $k > \frac{9}{2}$
B $k \geq \frac{9}{2}$
C $k > 9$
D $k < 0$ or $k > \frac{9}{2}$
E $0 < k < \frac{9}{2}$
F $k \neq 0$

Question 17

$OABC$ is the parallelogram with vertices $O(0, 0)$, $A(6, 0)$, $B(8, 4)$ and $C(2, 4)$. The point M lies on AB with $AM : MB = 1 : 2$, and the line OM meets the diagonal AC at P. What fraction of the area of the parallelogram is the area of triangle OPC?

A $\frac{1}{4}$
B $\frac{2}{7}$
C $\frac{1}{3}$
D $\frac{3}{8}$
E $\frac{2}{5}$
F $\frac{1}{2}$

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Question 18

The line $y = kx$, where $0 < k < 4$, divides the finite region enclosed by the curve $y = x(4 - x)$ and the x-axis into two parts of equal area. Find k.

A $4 - 2\sqrt{2}$
B $4 - 2\sqrt[3]{4}$
C 2
D $4 - \sqrt[3]{16}$
E $\frac{8}{3}$
F $4 - \sqrt[3]{2}$

Question 19

Find the complete set of real values of k for which both roots of

$$x^2 - 2kx + k + 6 = 0$$

are real and lie strictly between -3 and 3. (A repeated root counts as both roots.)

A no such k
B $-3 < k \leq -2$
C $-\frac{15}{7} < k < -2$
D $k \leq -2$
E $-\frac{15}{7} < k \leq -2$
F $-\frac{15}{7} < k < 3$

Question 20

For which values of the constant c is the point of the curve $y = x^2$ that is closest to $(0, c)$ the origin?

A $c \leq 0$
B $c < \frac{1}{2}$
C $c \leq 1$
D $c \leq 2$
E $c < 1$
F $c \leq \frac{1}{2}$

END OF PAPER

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